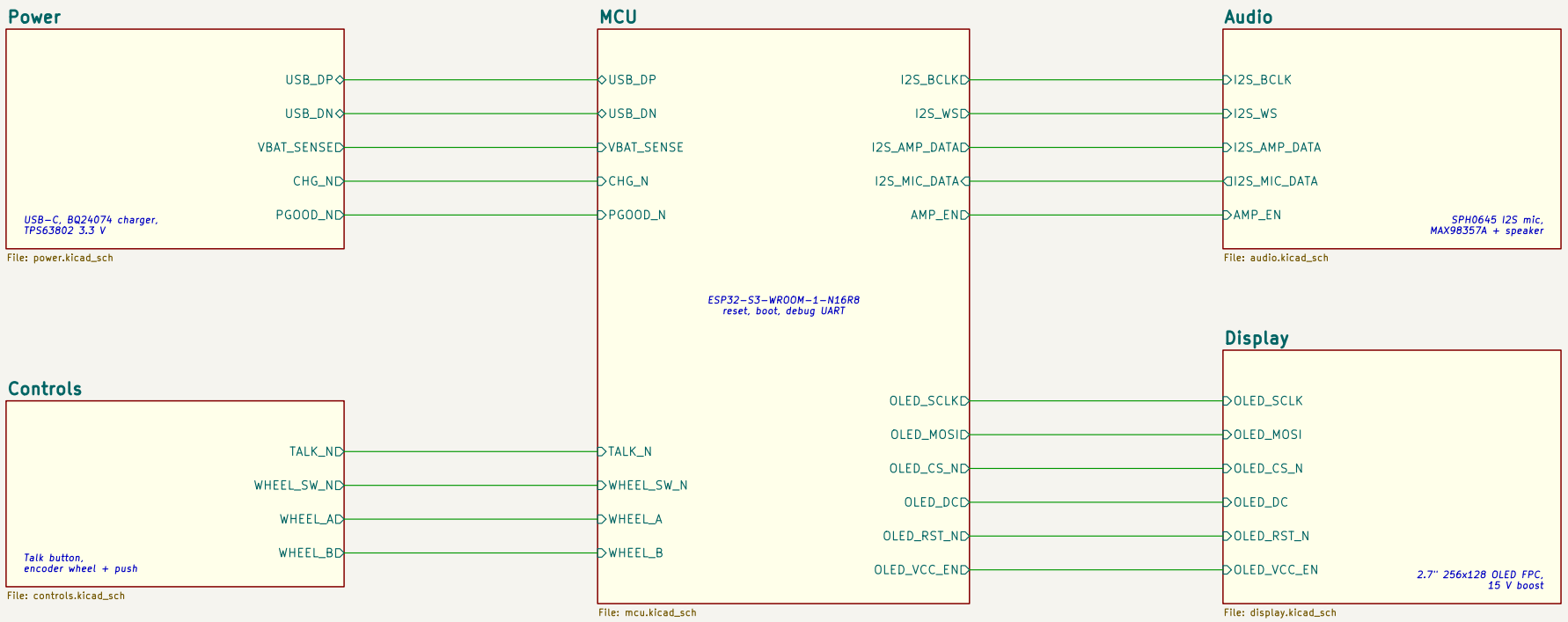
PocketBrain

System overview. Each block is one sheet. Power rails (VBUS, +BATT, VSYS, +3V3, +15V, GND) are global.



GPIO MAP

EN	Reset	RC + RESET button
IO0	BOOT	strapping, BOOT button
IO1	VBAT_SENSE	ADC1_CH0, VBAT / 2
IO2	CHG_N	low = charging
IO4	I2S_BCLK	mic + amp
IO5	I2S_WS	mic + amp
IO6	I2S_MIC_DATA	mic to MCU
IO7	I2S_AMP_DATA	MCU to amp
IO8	AMP_EN	new: PB_AMP_SD_PIN = 8
IO9	OLED_VCC_EN	new: 15 V boost enable
IO10	OLED_CS_N	
IO11	OLED_MOSI	
IO12	OLED_SCLK	
IO13	OLED_DC	
IO14	OLED_RST_N	
IO15	TALK_N	RTC wake
IO16	WHEEL_SW_N	RTC wake
IO17	WHEEL_A	
IO18	WHEEL_B	
IO19 / IO20	USB_DN / USB_DP	native USB
IO21	PGOOD_N	low = USB power present
IO43 / IO44	UART_TX / UART_RX	debug header

OPEN ITEMS

CHECK before PCB layout:

1. OLED flex on the TCC panel: pinout, length, contacts facing the display front.
2. Battery lead polarity (J2: pin 1 = cell +).
3. Fit-check Talk switch (TS-1101VS) and wheel (SIQ-02FVS3) in the shell; both sit on a board edge.
4. Wheel: B is undefined at the detent. Decode on A edges in firmware.
5. Firmware: drive IO8 and IO9, read IO1 / IO2 / IO21.

Schematic draft for bench bring-up; verify items marked CHECK

One-handed, battery-powered voice assistant

PocketBrain

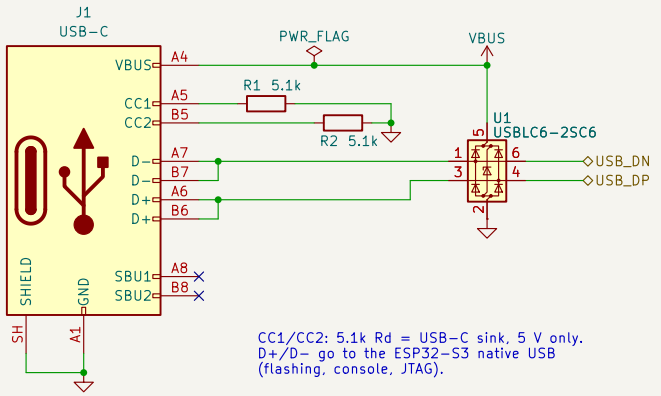
Sheet: /

File: pocketbrain.kicad_sch

Title: PocketBrain system overview

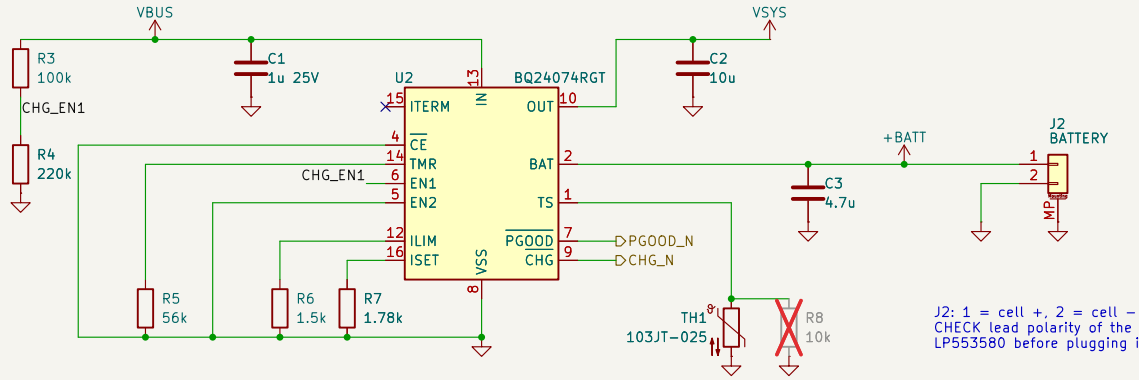
Size: A4	Date: 2026-11-11	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 1/6

USB-C INPUT



CC1/CC2: 5.1k Rd = USB-C sink, 5 V only.
D+/D- go to the ESP32-S3 native USB
(flashing, console, JTAG).

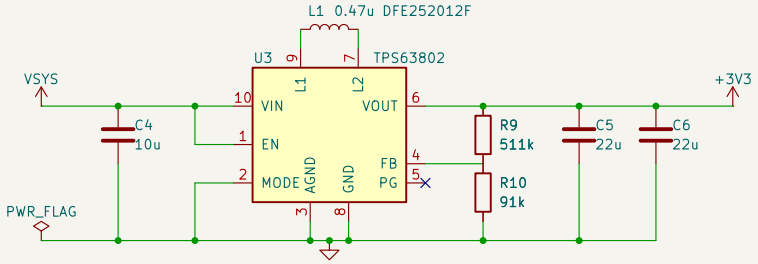
LI-ION CHARGER AND POWER PATH



J2: 1 = cell +, 2 = cell -
CHECK lead polarity of the
LP553580 before plugging in.

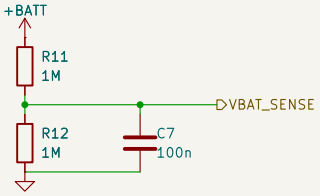
TH1: Semitec 103JT-025 film NTC (10k, B 3435), taped flat on the cell.
Charging is allowed from about 0 to 51 C. Fit R8 only to run without TH1.

3.3 V BUCK-BOOST



VOUT = 0.5 V x (1 + 511k/91k) = 3.31 V
MODE low = power-save (PFM) for deep sleep

BATTERY SENSE



VBAT / 2 to ADC1_CH0
2 uA divider current

NOTES

Charger (BQ24074)
- EN1 = 1 (while VBUS present), EN2 = 0: USB500 mode, input limited to 500 mA with VIN-DPM.
- Fast charge 890 / 1.78k = 0.5 A (0.28 C for 1800 mAh).
- Safety timers 56k: 45 min pre-charge, 7.5 h fast.
- ILIM 1.5k only used if EN2 is set high (1.07 A).
- ITERM open: charge ends at 10 % of the fast current.
- VSYS = 4.4 V on USB, VBAT on battery.
- CHG_N, PGOOD_N are open-drain; pull-ups are on the MCU sheet.

Rails
- VSYS: charger output, feeds 3V3, audio amp, OLED boost.
- +3V3: always on (deep sleep keeps RTC + wake GPIOs).

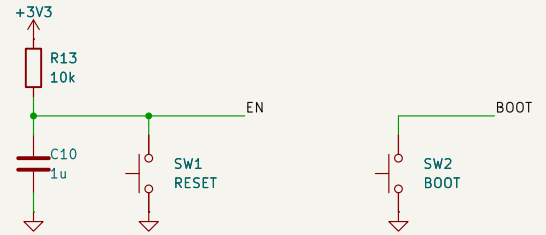
Schematic draft for bench bring-up; verify items marked CHECK
One-handed, battery-powered voice assistant
PocketBrain

Sheet: /Power/
File: power.kicad_sch

Title: Power: USB-C, charger, 3.3 V supply

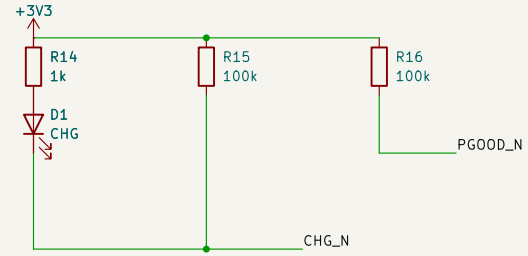
Size: A4	Date: 2026-11-11	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 2/6

RESET AND BOOT

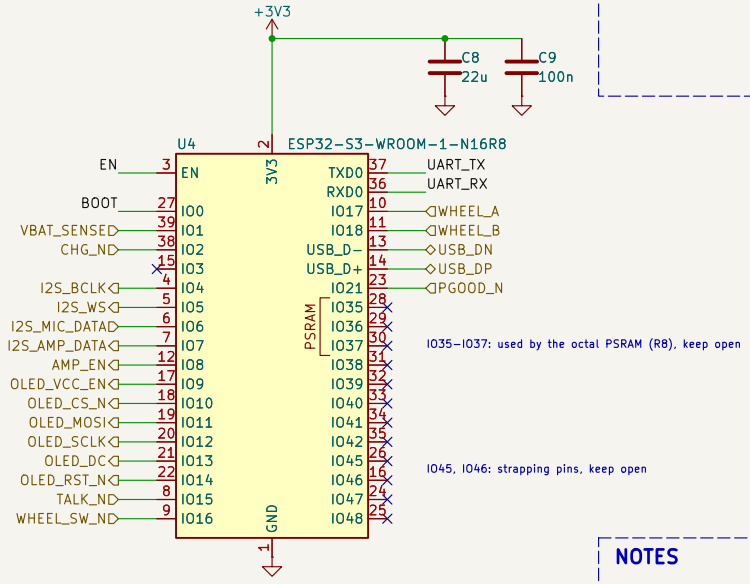


Hold BOOT (IO0 low) during reset for ROM download mode.

CHARGER STATUS



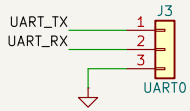
CHG_N low: charging (LED on)
PGOOD_N low: USB power present
Both open-drain from the BQ24074.



IO35–IO37: used by the octal PSRAM (R8), keep open

IO45, IO46: strapping pins, keep open

DEBUG UART



3.3 V logic, 115200 baud.
Optional: console is also on USB.

NOTES

- Module: 16 MB flash, 8 MB octal PSRAM.
- IO15 / IO16 are RTC GPIOs: ext1 wake sources (Talk, wheel push). IO21 (PGOOD_N) is RTC-capable too.
- GPIO numbers match embedded/main/Kconfig.projbuild, plus IO8 (AMP_EN), IO9 (OLED_VCC_EN). IO1, IO2, IO21.
- IO3, IO45, IO46 are strapping pins and stay open.
- Deep sleep: hold the OLED and I2S outputs (gpio_hold_en) so panel and microphone inputs never float.
- Keep the antenna end clear of copper, battery, speaker.

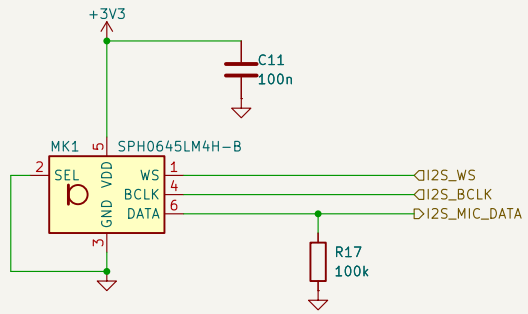
Schematic draft for bench bring-up; verify items marked CHECK
One-handed, battery-powered voice assistant
PocketBrain

Sheet: /MCU/
File: mcu.kicad_sch

Title: MCU: ESP32-S3-WROOM-1-N16R8

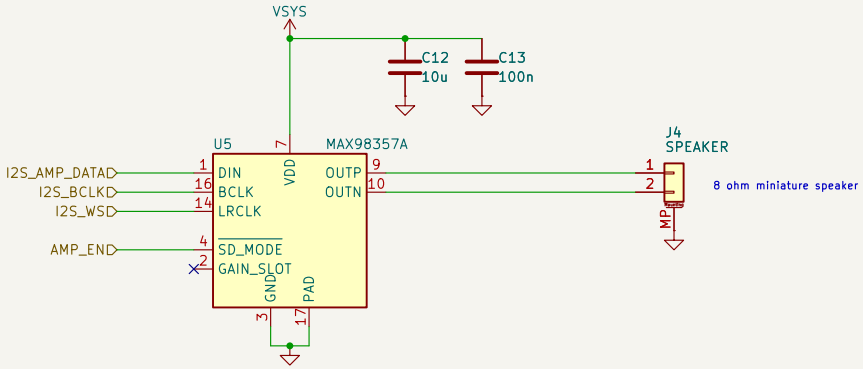
Size: A4	Date: 2026-11-11	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 3/6

I2S MEMS MICROPHONE



SEL low = left slot. DATA is tri-stated in the other slot, 100k keeps the line defined.
Bottom port: needs a PCB hole and a sealed acoustic path, away from the speaker.

CLASS-D SPEAKER AMPLIFIER



AMP_EN drives SD_MODE: low = shutdown (internal 100k pull-down), high (3.3 V) = on, left slot. GAIN open = 9 dB.
Supplied from VSYS (3.4-4.4 V) to keep speaker peaks off +3V3.
Filterless output: keep the speaker leads short and twisted.

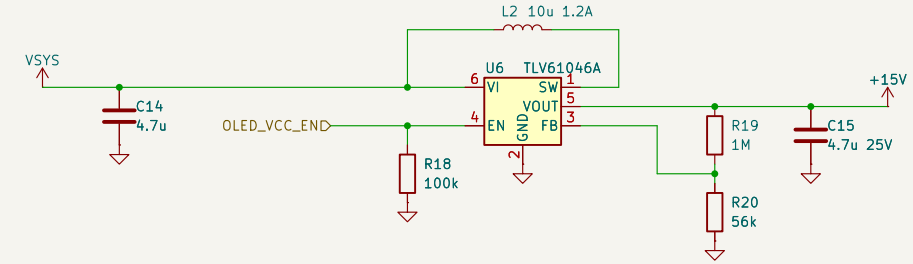
Schematic draft for bench bring-up; verify items marked CHECK
One-handed, battery-powered voice assistant
PocketBrain

Sheet: /Audio/
File: audio.kicad_sch

Title: Audio: I2S microphone and class-D amplifier

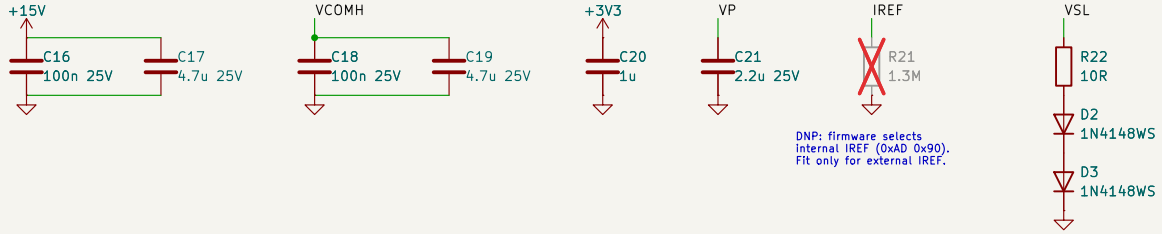
Size: A4	Date: 2026-11-11	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 4/6

OLED PANEL SUPPLY: 15 V BOOST



VCC = 0.795 V x (1 + 1M/56k) = 15.0 V (panel: 14.5–15.5 V)
EN low: output disconnected, VCC floats as the panel requires.
Sequence: +3V3 and init first, then EN high, 100 ms, display on.
Panel draws about 53 mA at 50 % pixels on, 91 mA at 100 %.

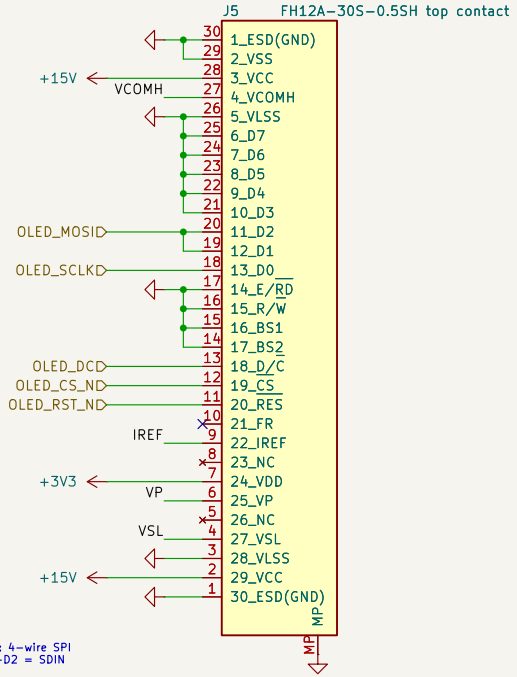
PANEL SUPPORT PARTS (SSD1363 4-wire SPI reference circuit)



DNP: firmware selects internal IREF (0xAD 0x90). Fit only for external IREF.

Place all of these next to the FPC connector. 25 V parts on VCC, VCOMH and VP.

OLED FPC



BS1 = BS2 = 0: 4-wire SPI
D0 = SCLK, D1+D2 = SDIN

Pin names start with the FPC pin number. Numbers outside the box are J5 pads: the tail folds behind the board, so pad n mates FPC pin 31 − n.

CHECK on the TCC panel:
- pinout and flex length
- contacts face the display front (Midas drawing), so J5 must be TOP contact

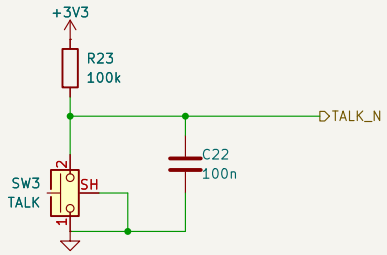
Schematic draft for bench bring-up; verify items marked CHECK
One-handed, battery-powered voice assistant
PocketBrain

Sheet: /Display/
File: display.kicad_sch

Title: Display: 2.7 in 256x128 OLED (SSD1363) and 15 V supply

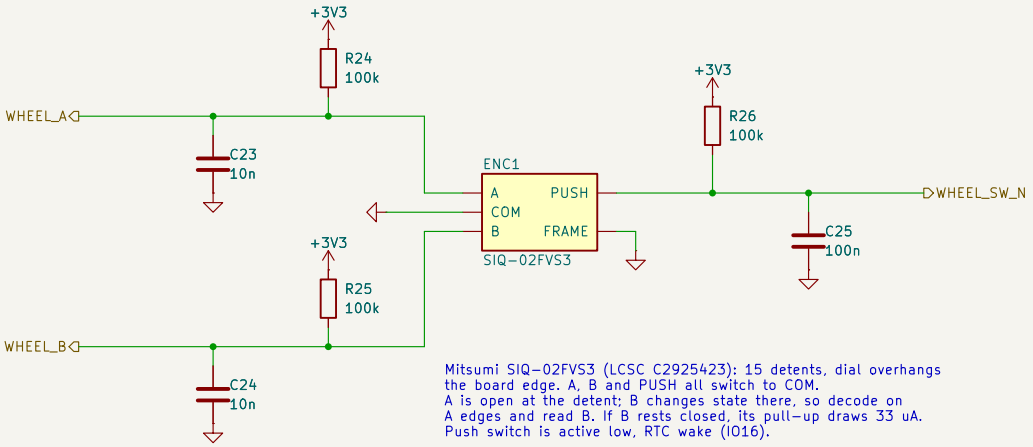
Size: A4	Date: 2026-11-11	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 5/6

TALK BUTTON



XKB TS-1101VS-B-A-A (LCSC C561499): side push, 160 gf, metal cover grounded. Sits on the board edge, recessed, opposite the wheel. Active low, RTC wake (IO15).

SCROLLWHEEL: DETENTED ENCODER WITH PUSH SWITCH



Mitsumi SIQ-02FVS3 (LCSC C2925423): 15 detents, dial overhangs the board edge. A, B and PUSH all switch to COM.
A is open at the detent; B changes state there, so decode on A edges and read B. If B rests closed, its pull-up draws 33 uA.
Push switch is active low, RTC wake (IO16).

Schematic draft for bench bring-up; verify items marked CHECK
One-handed, battery-powered voice assistant
PocketBrain

Sheet: /Controls/
File: controls.kicad_sch

Title: Controls: Talk button and scrollwheel

Size: A4	Date: 2026-11-11	**Rev: 0.3**
KiCad E.D.A. 10.0.8		Id: 6/6